

 GRADUATION PROJECT

Complaints Management System

An institutional platform to receive, track, examine and respond to citizen complaints — with OCR intake, AI triage, social media monitoring and powerful reporting.

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— REACT · NESTJS · POSTGRESQL —

What we will cover

1 Introduction

2 Objectives

3 AI & Multi-Agent Objectives

4 Technologies Used

5 System Features

6 System Architecture

7 AI & Multi-Agents

8 Advanced RAG & Evaluation

9 Frontend Walkthrough

10 Testing

11 System Demonstration

12 Future Enhancements

13 Conclusion

Institutional Complaints Management

A **full-stack system** that helps organizations manage the entire citizen complaint workflow — intake, examination, department routing, responses, letters and archiving — via a **secure, role-based** web platform enhanced with **OCR** and **AI**.

📋 Complaints

👤 Citizens

🏢 Departments

📄 Letters

📊 Reports

🗣️ Social Media

What we aim to achieve

Digitizing the full complaint lifecycle with secure roles, AI-assisted intake, reporting and monitoring.



Digitize the Lifecycle

Move the full complaint flow — intake to response to archive — online.



Secure Role-based Access

JWT authentication for Super Admin, Admin and Official roles.



Reporting & Analytics

KPIs, custom builders, scheduled reports and Excel / PDF exports.



Letters & Memos

Generate official letters and government memos from templates.

One system that does **the heavy lifting**

A Mastra multi-agent core automates intake, OCR, triage, scraping and drafting
— so officials only review and approve.



Zero-Typing Intake

The OCR Intake Agent extracts structured data straight from scanned complaint forms.



Smart Triage & Routing

The Complaint Triage Agent flags severity, detects duplicates and routes each case to the right department.



Social Monitoring Agent

The Social Intake Agent watches Facebook groups and turns new posts into complaint drafts automatically.



Instant Summaries

The Social Summary Agent condenses threads and the Summarization module produces daily reports.



Less Manual Work

Mastra orchestrates all five agents — intake, OCR, triage, scraping and drafting happen end-to-end, automatically.

The tech stack



REACT

Frontend

✓ React 19 + TypeScript

✓ Vite + Tailwind CSS v4

✓ React Router · TanStack Query

✓ shadcn/ui + Base UI · Radix

✓ Recharts · Zod · React Hook Form

✓ Zustand · Axios · Jodit editor



NESTJS

Backend

✓ NestJS (Express) · JWT + Passport

✓ Prisma ORM · class-validator

✓ Tesseract OCR · Mammoth (DOCX)

✓ ExcelJS · Puppeteer · Apify

✓ node-cron · bcrypt



POSTGRESQL

Database

✓ PostgreSQL

✓ Prisma Migrations & Seed

✓ pgvector · vector search



MASTRA + GEMINI

AI & Multi-Agents

✓ Mastra Framework · multi-agent orchestration

✓ Google Gemini 2.5 Flash + embeddings

✓ pgvector semantic search (RAG)

✓ Structured output (Zod schemas)

✓ Advanced RAG retrieval techniques

✓ RAG triad evaluation

Nine core modules



Authentication & Roles

JWT login for Super Admin, Admin & Official.



Complaint Registration

Citizens, types, severity, reception & attachments.



OCR Intake

Scan documents to auto-fill complaints.



AI Triage & Summarization

Duplicate detection, severity analysis & summaries.



Department Routing

Multi-department assignment, responses & deadlines.



Social Media Monitoring

Scraped groups, drafts & polls.



Letter Templates

Placeholders, DOCX import & template builder.



Official Letters & Memos

Generate official letters & memos with PDF export.



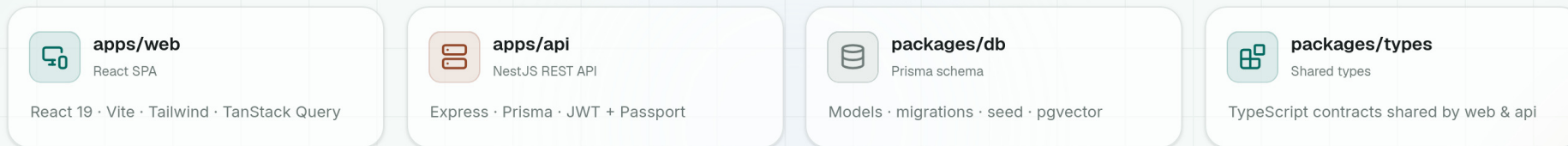
Reporting & Analytics

KPIs, custom & scheduled reports, Excel export.

A modular monorepo, **fully connected**

Two apps and two shared packages, connected through one REST API, a pgvector-backed database and an orchestrated AI layer.

MONOREPO STRUCTURE



REQUEST FLOW



One framework, **many specialists**

Mastra Framework orchestrates dedicated agents — each with a single, focused job and structured, validated output.



Complaint Triage Agent

Severity assessment & recurrence detection via semantic similarity.



OCR Intake Agent

Extracts structured citizen & complaint fields from scanned Arabic forms.



Social Intake Agent

Filters spam and extracts complaint fields from Facebook posts.



Social Summary Agent

Generates concise overviews of the complaints captured by each poll.



ComplaintsAgent

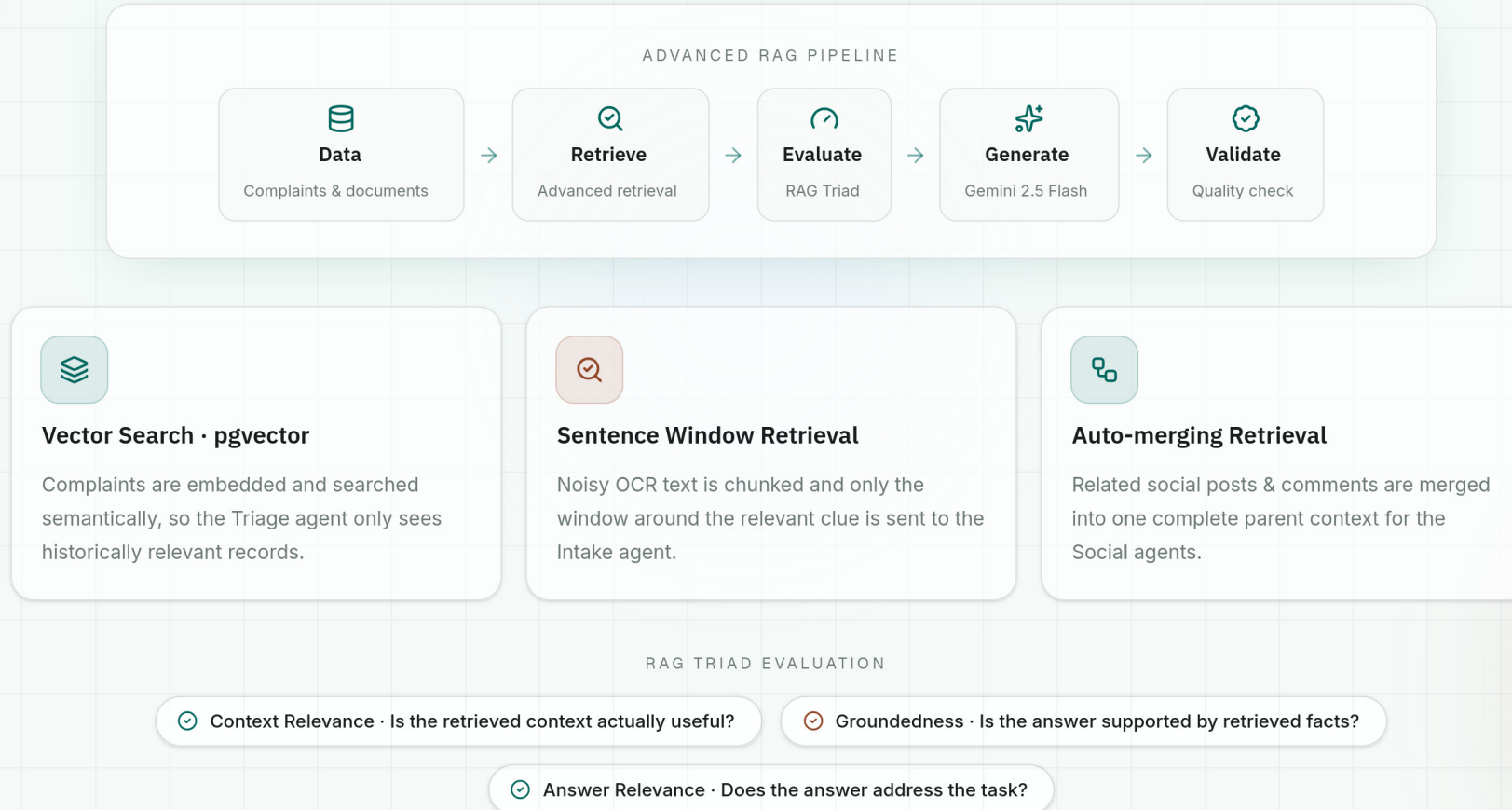
Writes Arabic summaries, official reports and government memos.

ORCHESTRATION

Mastra routes every call through one model router — **Google Gemini 2.5 Flash** — with **Zod structured outputs** and **pgvector semantic memory**.

From raw prompts to **controlled context**

The course solution upgrades direct LLM calls into an advanced RAG pipeline with retrieval and evaluation.



Twelve **pages**, one flow

A



Login

B



Dashboard Page

C



Complaints List Page

D



New Complaint Page

E



Complaint Detail Page

F



OCR Intake Page

G



Due Assignments Page

H



Departments & Reference Data

I



Letter Templates Page

J



Letters & Memos Page

K



Users Page

L

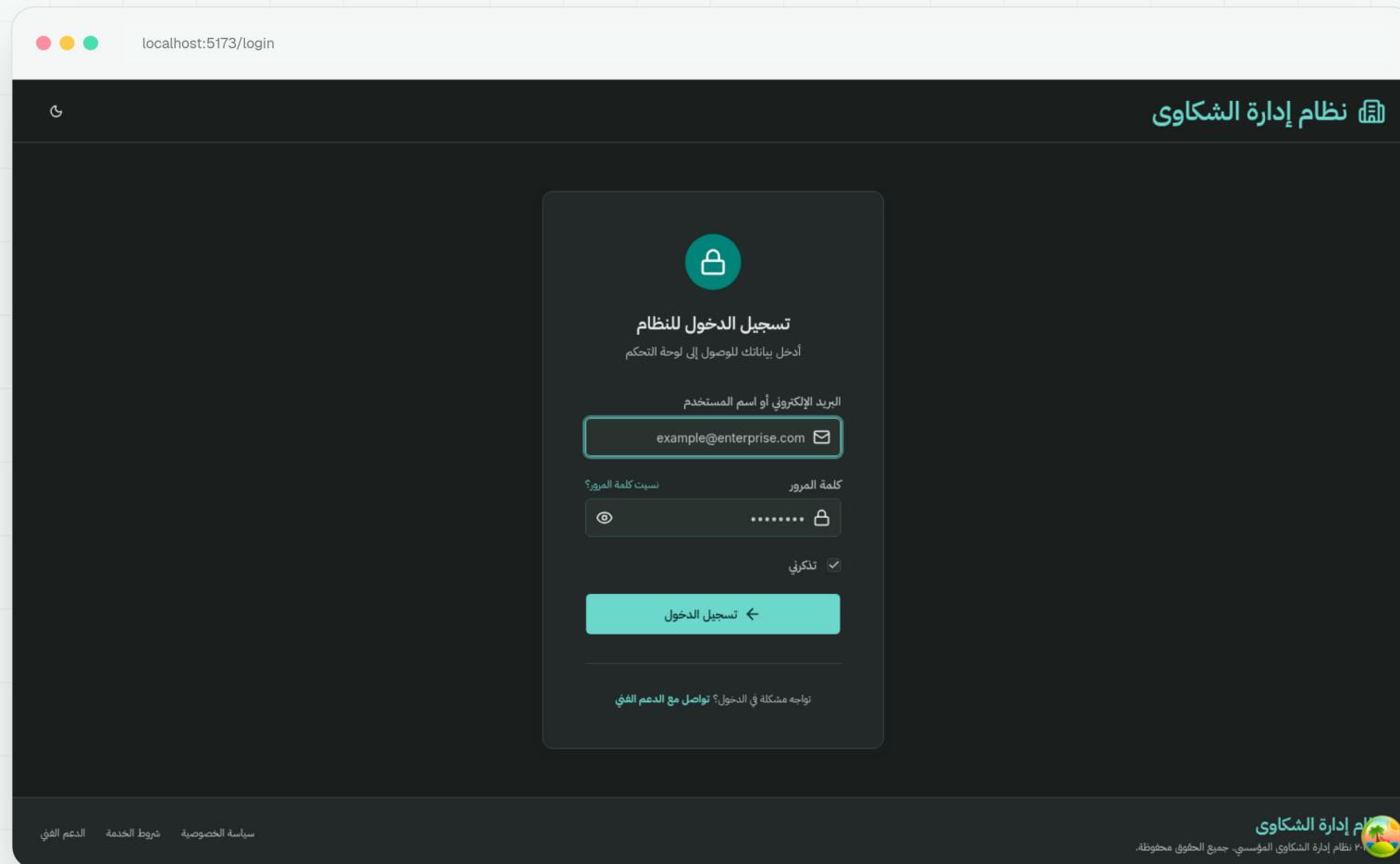


Social Monitoring Page

I. Login

Secure sign-in with role-based access (Super Admin / Admin / Official). New users are created by admins from the dashboard's Users page, not via self-registration.

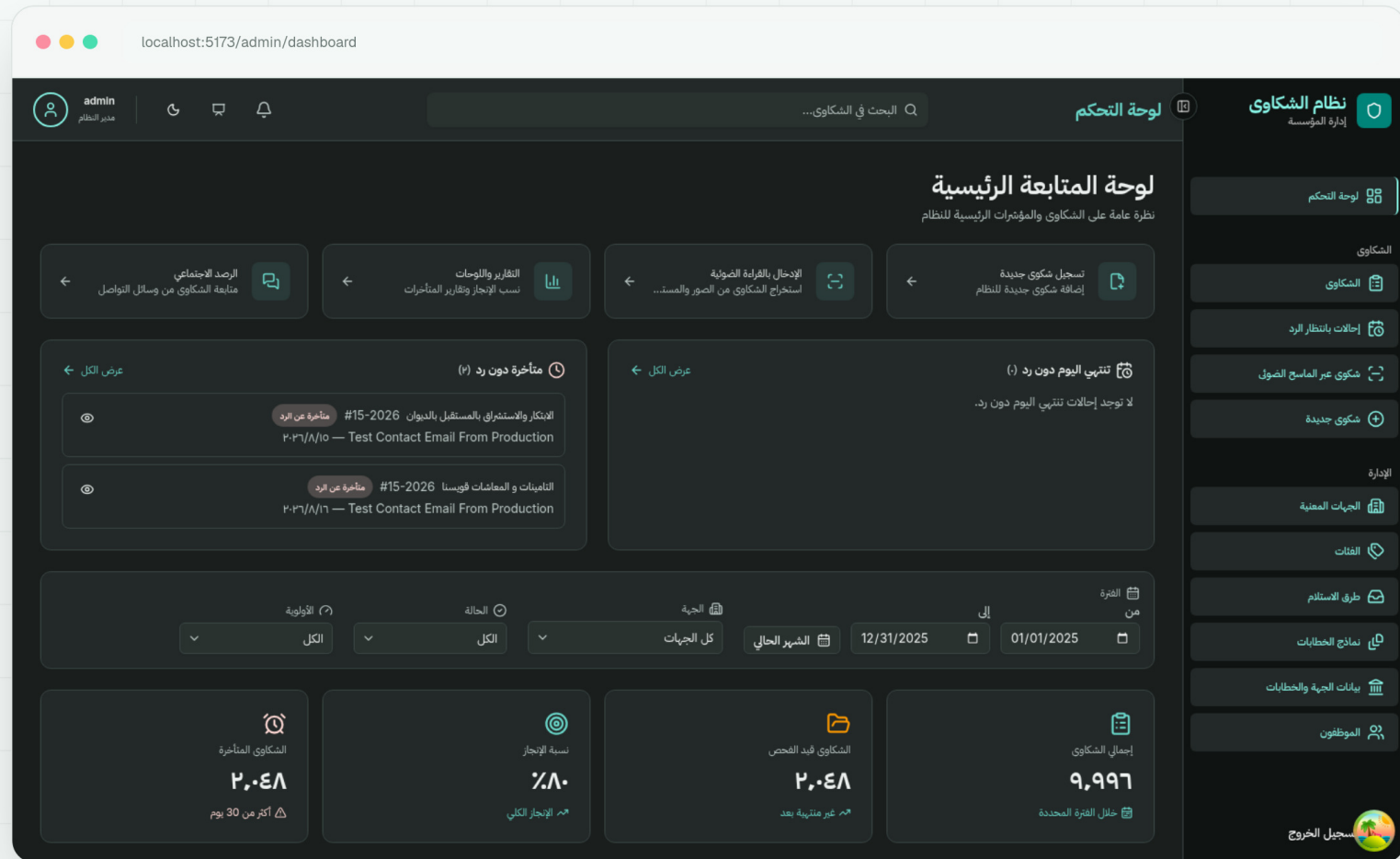
- JWT authentication for all roles
- Users created by admins from dashboard
- Role-aware redirects



II. Dashboard

Provides an overview of the system: total complaints, open & delayed items, resolution KPIs, quick actions, a status donut and delay analytics.

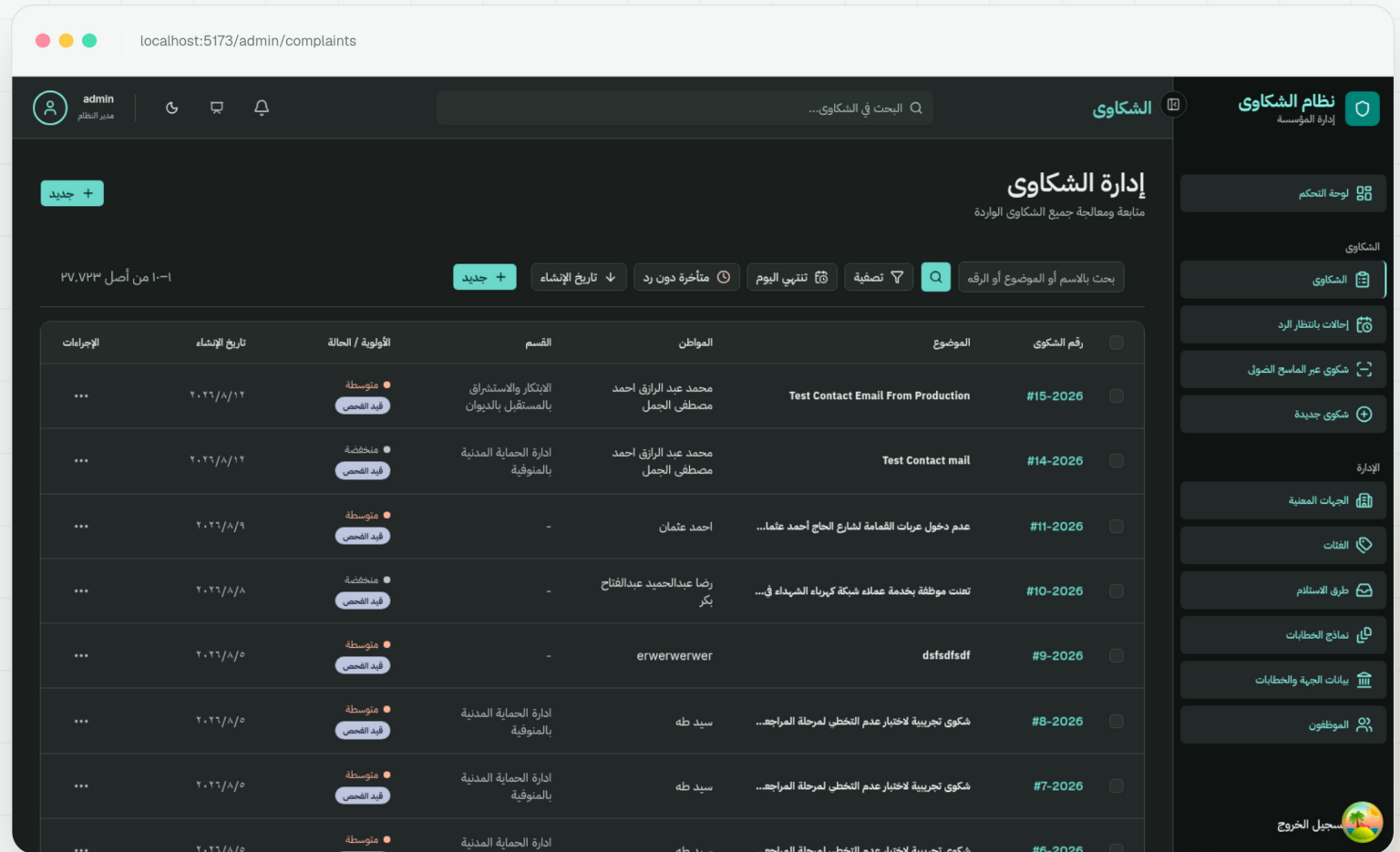
- Complaints & status counters
- Delays and achievement analytics
- Quick action shortcuts



III. Complaints List

Displays complaints with powerful filtering and search, batch actions, and card or table views with full status visibility.

- Filter by department, type & status
- Search and batch AI actions
- Table and card views



IV. New Complaint

A guided multi-step wizard that captures citizen details, complaint info, attachments and a final review before submission.

- Multi-step wizard flow
- Citizen lookup by national ID
- Attachments & review step

localhost:5173/admin/complaints/new

admin
مدير النظام

🔍 البحث في الشكاوى...

شكوى جديدة

إدارة المؤسسة

لوحة التحكم

الشكاوى

الشكاوى

إحالات بانتظار الرد

شكوى عبر الماسح الضوئي

شكوى جديدة

الإدارة

الجهات المعنية

الوثائق

طرق الاستلام

نماذج الخطابات

بيانات الجهة والخطابات

الموظفون

سجل الخروج

إدخال عبر الماسح الضوئي

تقديم شكوى جديدة

يرجى تعبئة التفاصيل أدناه لمساعدتنا في معالجة شكاواك بفعالية.

1 بيانات المواطن

2 بيانات الشكوى

3 المرفقات

4 المراجعة

معلومات المواطن

الرقم القومي

رقم الجوال

الرقم القومي

الرقم الجوال (اختياري)

الاسم الكامل *

الاسم الكامل للمواطن

العنوان

العنوان (اختياري)

القرية

المركز

اختر المركز أولاً

اختر المركز

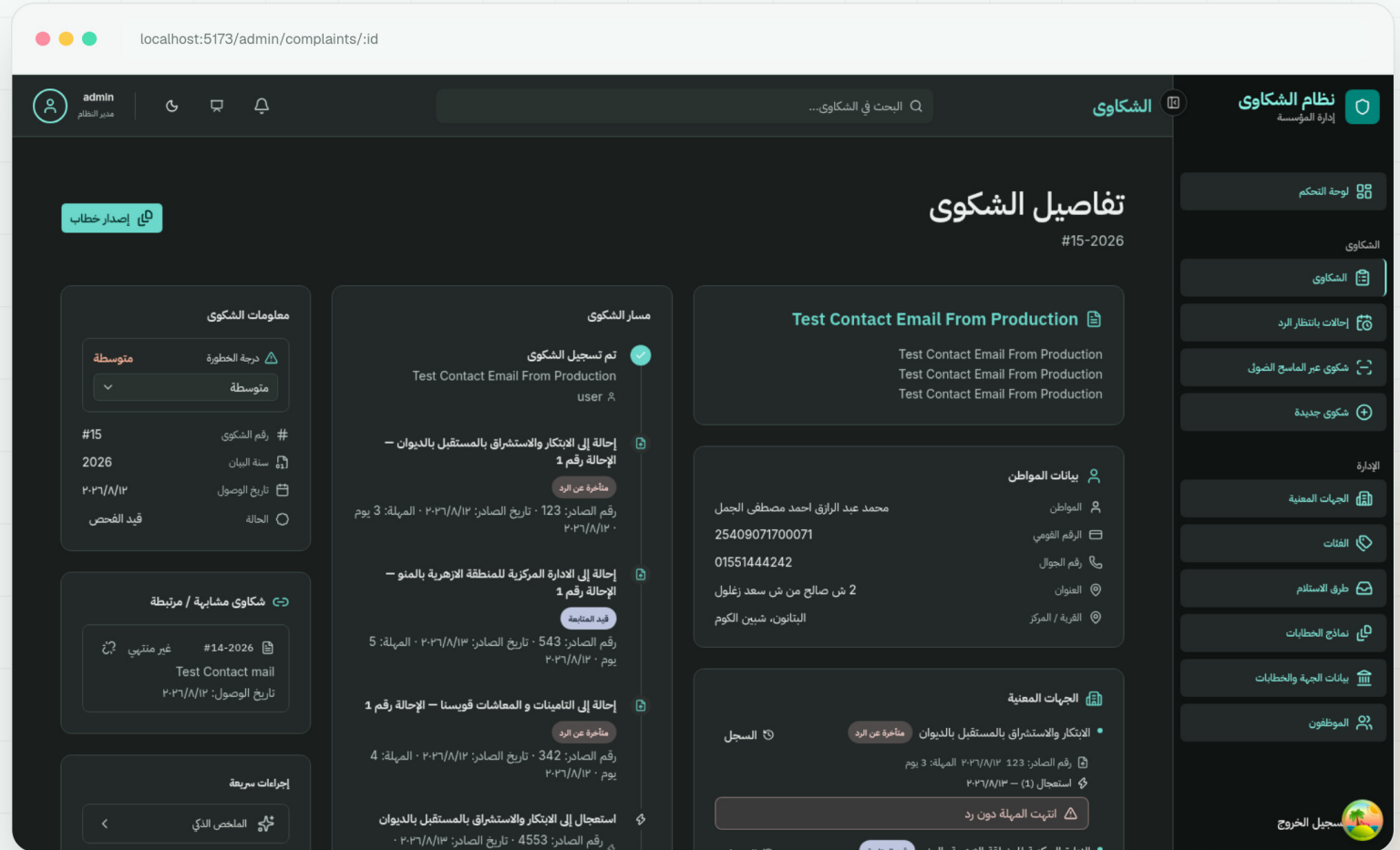
شكاوى المواطن السابقة

اعرض الشكاوى المسجلة مسبقاً لنفس الرقم القومي أو بنفس الاسم لتجنب تكرار التسجيل.

V. Complaint Detail

A rich detail page with timeline, examination status, department responses, evidence gallery, related links and quick actions.

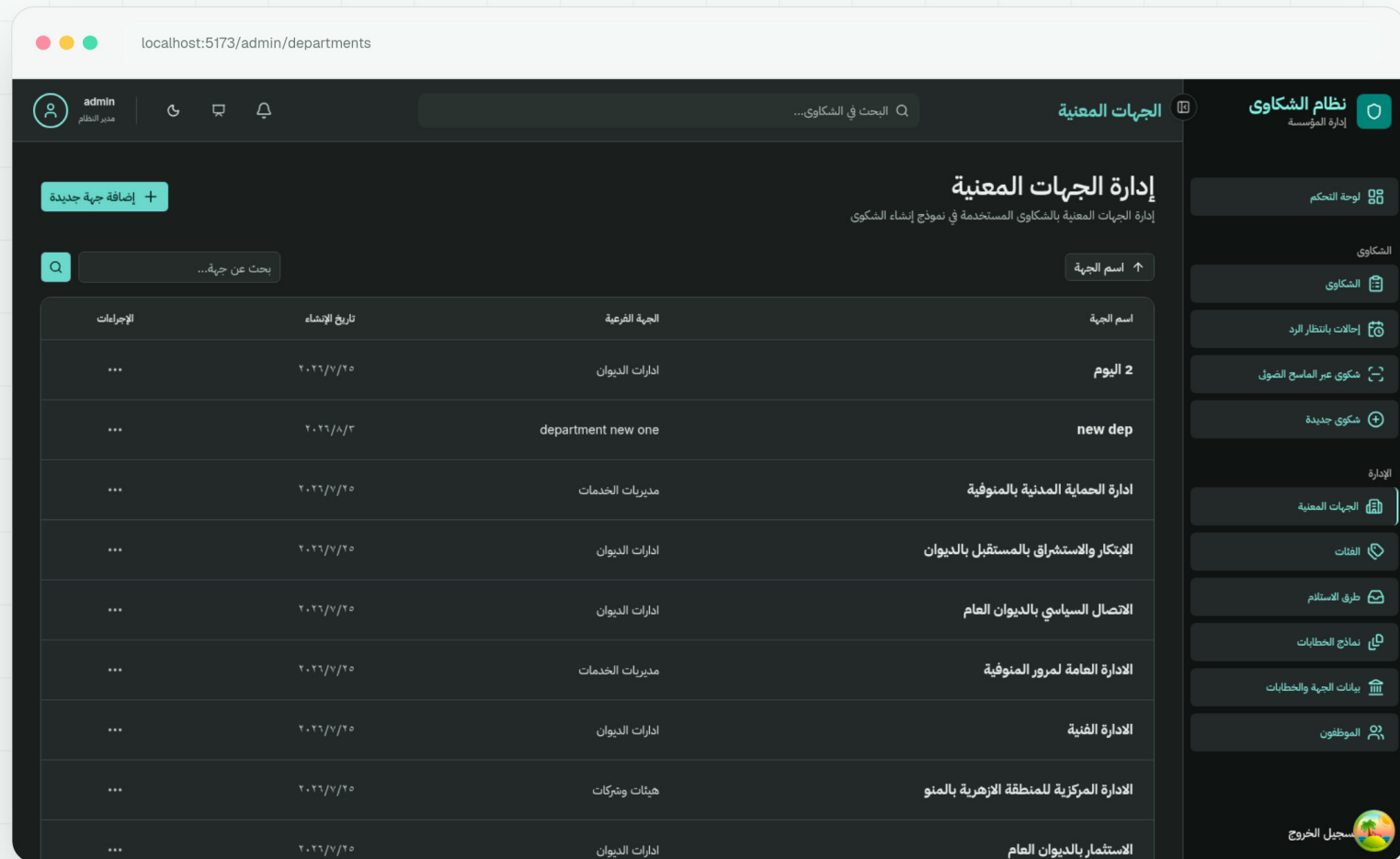
- Lifecycle timeline & statuses
- Evidence gallery & attachments
- Responses, urgency & actions



VI. Departments & Reference Data

Manage departments, complaint types and reception methods that power the intake forms and routing workflow.

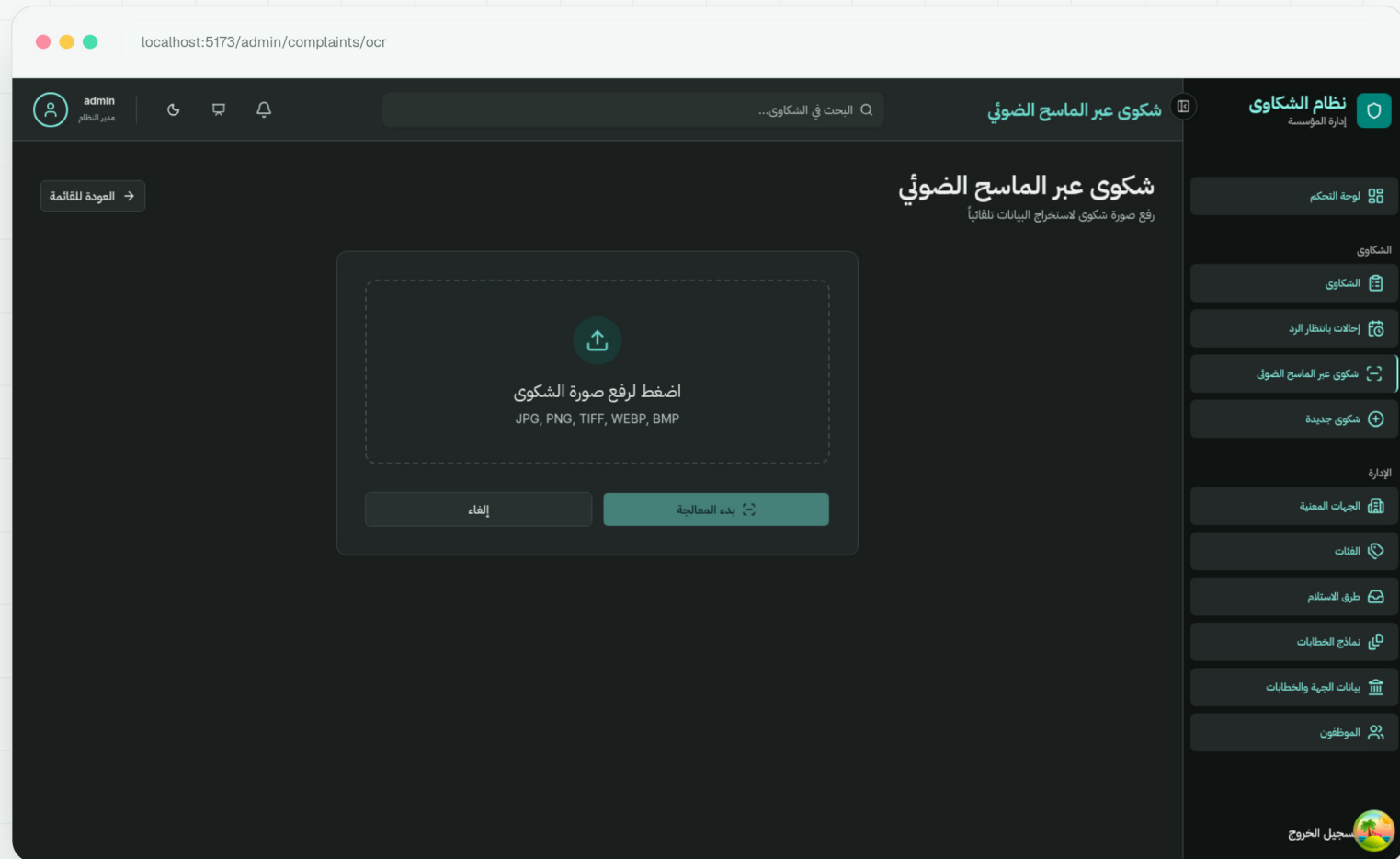
- Departments management
- Complaint types & reception methods
- Simple CRUD interfaces



VII. OCR Intake

Scan a paper complaint document and let OCR auto-extract citizen data and complaint details into the form.

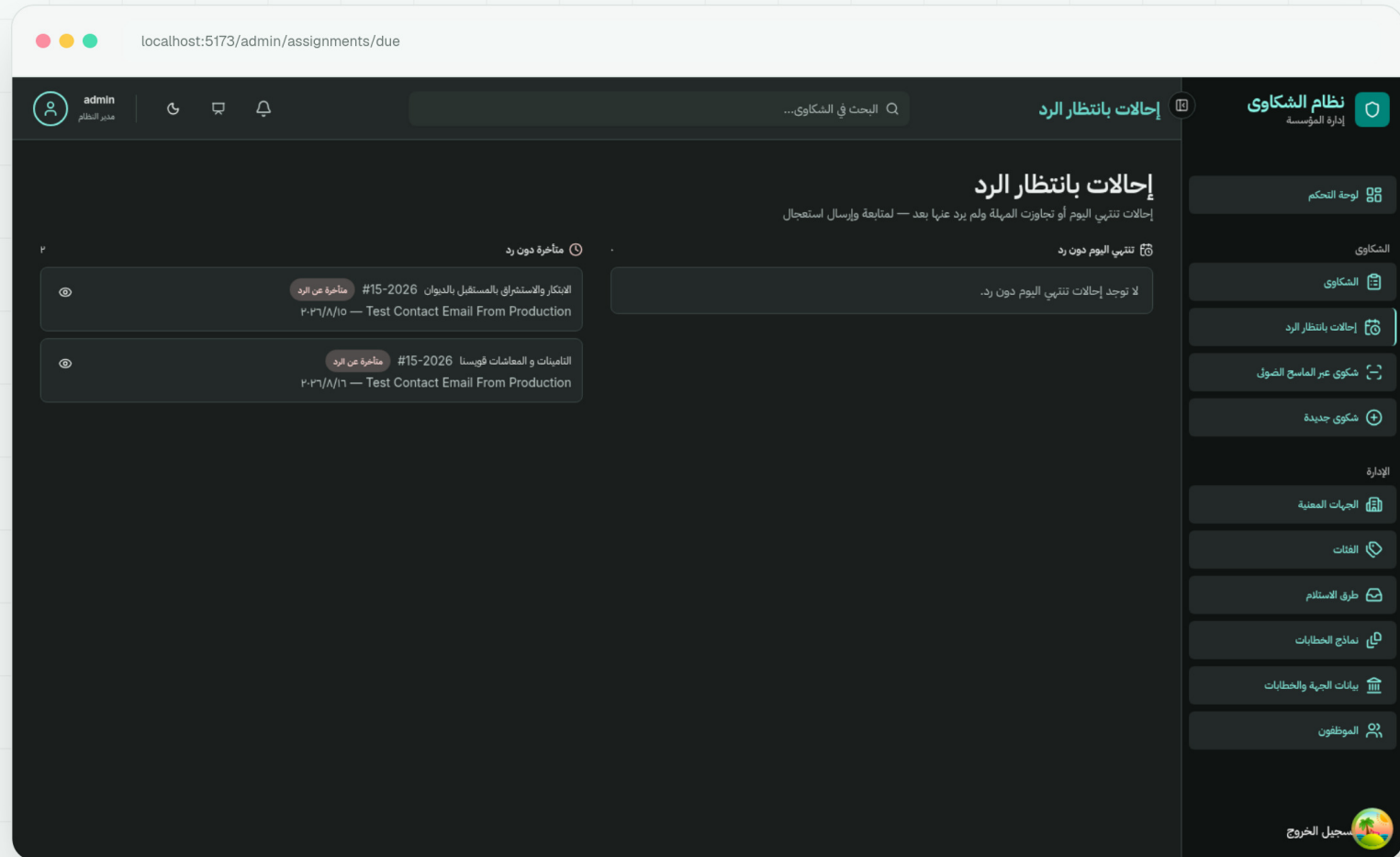
- Tesseract-based OCR extraction
- Auto-fill citizen & subject fields
- Fast document intake



VIII. Due Assignments

Lists complaints assigned to the current user that are awaiting response, with response deadlines and status tracking.

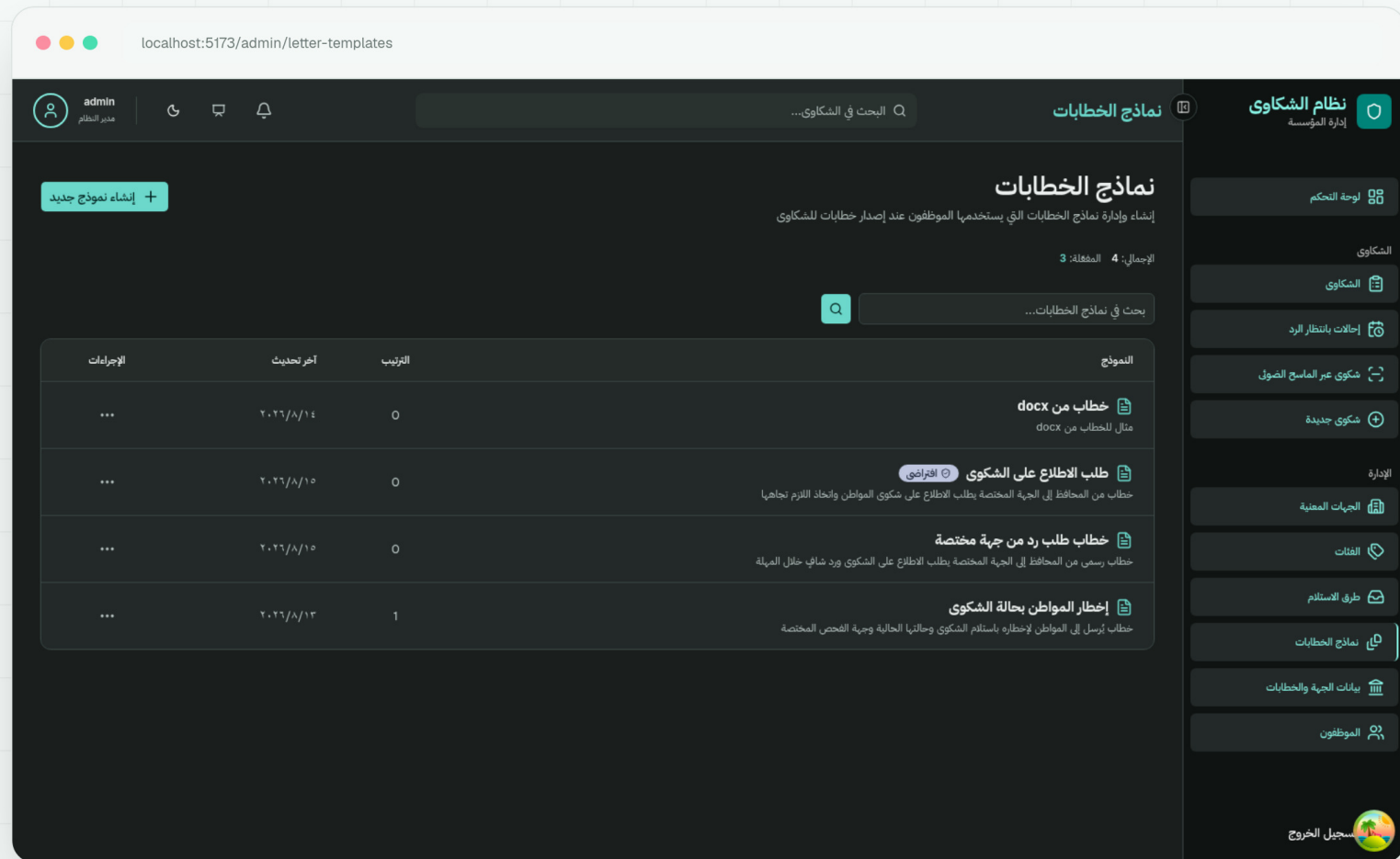
- Awaiting-response list
- Deadline progress indicators
- Respond directly from the list



IX. Letter Templates

Create official letter templates with placeholders, import from DOCX, preview drafts and generate letters and PDFs.

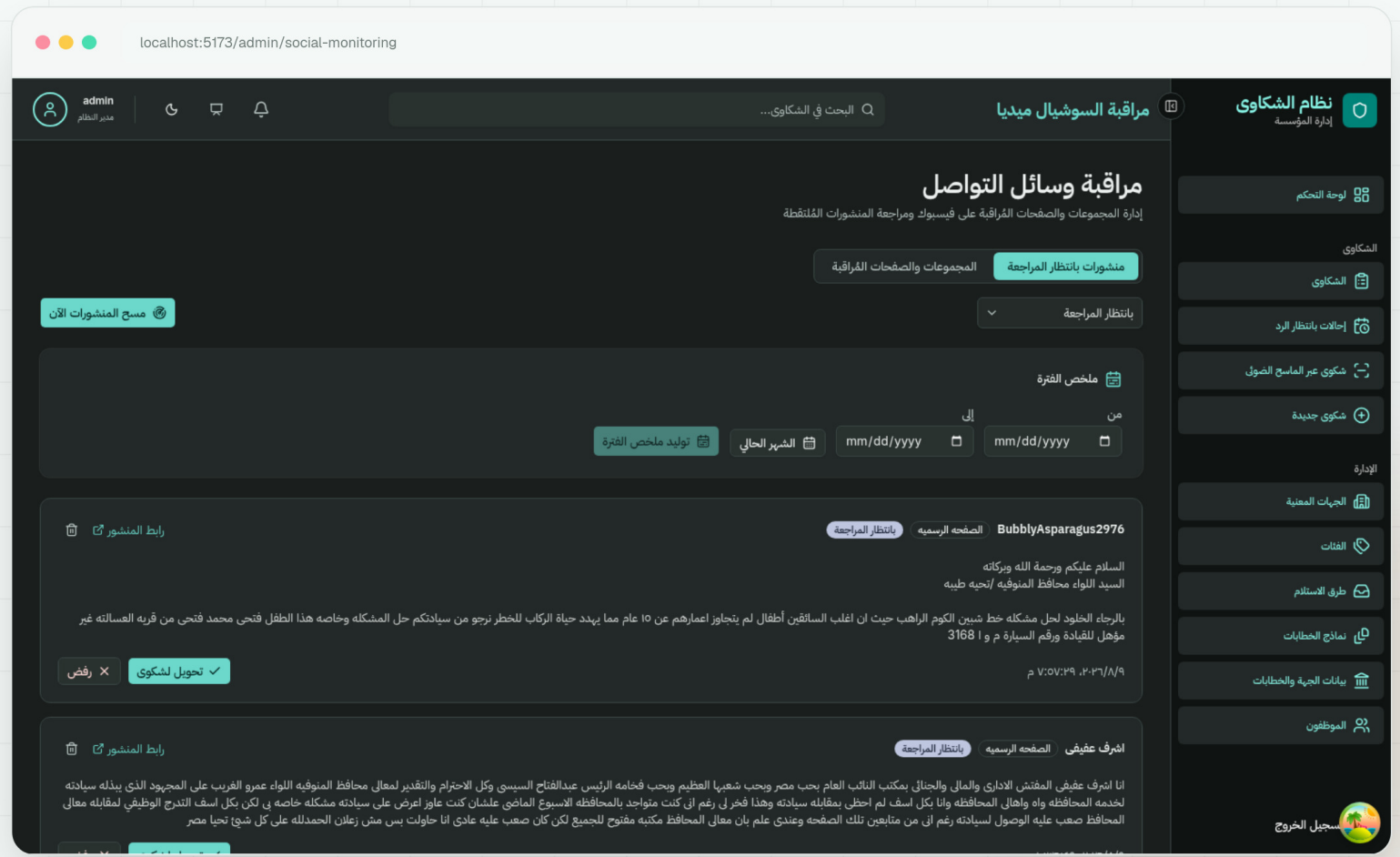
- Template builder with placeholders
- DOCX import & live preview
- Generate official letters / PDFs



X. Social Media Monitoring

Monitors scraped social groups for potential complaints, keeps drafts, and links relevant posts to complaint records.

- Scrape monitored groups
- Review & link drafts to complaints
- Group enable / disable controls



Quality, automated

Every push and pull request is verified by an automated CI pipeline before it reaches the reviewers.



CI Pipeline

GitHub Actions runs typecheck, Prisma migrations & seed, the API build and automated tests on every push and PR.



API Tests (Bruno)

End-to-end endpoint collections for Auth, Users, Complaints, Letters, Reports, AI, Intake and Social — run against a live pgvector PostgreSQL.



Type Safety

Strict TypeScript typecheck across the web app and the NestJS API catches errors before they ship.



Database Validation

Prisma schema migrations are deployed and the database is seeded with pgvector enabled in CI.

See it in action



Full walkthrough video

Live demo of the Complaints Management System end to end.

 [Video Link](#)

The road ahead

1



Mobile App (React Native / Flutter)

2



Email / SMS Notifications

3



AI Chatbot Assistant with NLP

4



Advanced Charts & Dashboards

5



Deployment on Docker + Cloud (AWS)

6



Citizen Self-Service Portal

7



Multi-Language UI (AR / EN)

8



More Report Exports (Excel / PDF)

9



Audit Trail & Soft-Delete Recovery

A modern, **scalable** complaints platform

The Complaints Management System provides a **modern, scalable solution** for handling institutional complaint workflows. Built on **React, NestJS and PostgreSQL**, it includes OCR intake, AI triage & summarization, social media monitoring, letter generation and an intuitive interface tailored for government workflows.

📄 OCR Intake

🧠 AI Triage & Summaries

🗣️ Social Monitoring

📊 Reporting

Useful links

All source code and documentation live in the GitHub repository — docs will be published under docs/ and linked here.



GitHub Repository

github.com/OS46-GP/complaints-syst...

Open



API Documentation

docs/ · coming soon

Soon



Trello

Project board

Soon



Figma

UI designs · ITI-GP-Complaints-System

Open

✦ QUESTIONS?

Thank You!

Feel free to ask any questions.

MH

Mahmoud Halim

MN

Mohamed Nassar

MI

Mahmoud Ismail

MS

Mostafa Safwat

AW

Amir Whdan